

SONOMA COUNTY AP, CA, USA

WMO: 724957

Lat: 38.504N

Lon: 122.810W

Elev: 114

StdP: 14.64

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23213

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	29.3	31.5	18.9	14.3	57.3	24.3	18.5	51.0	21.1	55.2	18.2	54.3	1.6	300	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.9	94.9	66.5	91.0	65.7	87.3	64.5	69.0	89.9	67.4	87.3	65.9	84.3	8.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	80.3	74.7	59.3	75.6	72.6	58.0	72.2	71.0	33.1	90.5	31.7	87.5	30.6	84.4	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.0	12.6	DB	25.3	101.7	2.3	3.3	23.6	104.0	22.3	105.9	21.0	107.7	19.4	110.1	(4)
(5)			WB	24.4	72.9	2.5	3.6	22.6	75.5	21.1	77.6	19.8	79.6	18.0	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	47.8	50.3	53.3	56.0	60.8	65.8	67.4	67.6	66.2	61.1	53.0	47.5	(6)
(7)		DBStd	8.86	4.41	4.86	5.16	5.49	5.02	5.18	4.09	3.86	4.92	5.08	5.27	5.04	(7)
(8)		HDD50	319	97	50	25	9	0	0	0	0	0	0	28	110	(8)
(9)		HDD65	2901	535	412	364	274	150	48	17	15	39	144	360	543	(9)
(10)		CDD50	3279	27	58	126	190	336	475	541	545	487	343	119	32	(10)
(11)		CDD65	386	0	0	1	5	20	73	93	95	76	22	1	0	(11)
(12)		CDH74	7287	3	11	69	214	572	1421	1464	1447	1401	640	44	1	(12)
(13)		CDH80	3069	0	1	11	60	203	633	621	639	659	237	5	0	(13)
(14)	Wind	WSAvg	4.6	3.3	4.5	5.0	5.5	5.8	5.7	5.4	4.9	4.3	3.9	3.3	3.6	(14)
(15)	Precipitation	PrecAvg	45.4	9.8	8.4	5.4	3.0	1.9	0.4	0.0	0.1	0.1	2.2	4.2	10.2	(15)
(16)		PrecMax	71.9	30.0	25.3	18.8	9.3	6.9	1.4	0.1	1.2	0.4	6.0	11.1	23.1	(16)
(17)		PrecMin	24.8	0.9	0.4	0.2	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.3	2.4	(17)
(18)		PrecStd	13.6	6.9	5.7	5.3	2.3	2.1	0.4	0.0	0.3	0.1	1.5	3.1	6.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.2	75.5	81.1	87.1	92.1	98.2	98.7	97.6	98.3	93.0	79.2	67.8	(19)
(20)			MCWB	52.5	58.9	61.8	63.8	65.9	68.2	70.1	67.8	66.7	62.5	60.6	52.4	(20)
(21)		2%	DB	64.0	69.3	74.9	80.3	85.6	92.4	91.6	92.2	93.5	86.4	73.8	62.9	(21)
(22)			MCWB	52.5	56.2	58.8	61.5	62.9	66.4	67.4	66.6	65.2	60.9	57.5	52.1	(22)
(23)		5%	DB	60.1	64.7	70.2	74.4	80.5	87.5	86.8	87.2	88.2	81.3	68.7	59.8	(23)
(24)			MCWB	51.9	53.1	56.1	58.7	61.6	64.9	66.2	65.7	63.9	59.1	55.7	53.1	(24)
(25)		10%	DB	56.9	60.5	65.4	69.2	75.3	82.2	82.2	82.2	82.2	75.9	64.0	57.3	(25)
(26)			MCWB	51.4	51.8	54.1	56.2	59.2	63.2	64.8	64.6	62.7	58.1	54.3	52.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	60.3	63.6	66.3	68.4	71.1	71.3	70.5	69.8	66.0	63.3	59.2	(27)
(28)			MADB	62.1	71.3	78.1	82.3	87.5	93.7	94.5	92.5	91.4	82.5	75.6	61.4	(28)
(29)		2%	WB	55.7	57.8	60.2	62.5	65.1	68.2	68.4	67.9	67.1	63.5	60.3	57.3	(29)
(30)			MADB	59.4	65.8	72.6	77.6	82.5	89.1	89.1	88.5	88.1	80.9	68.5	59.2	(30)
(31)		5%	WB	54.2	55.5	57.8	59.8	62.5	65.9	66.6	66.2	65.2	61.6	58.2	55.3	(31)
(32)			MADB	57.1	61.4	67.1	72.4	77.7	84.7	85.2	85.1	84.2	75.1	64.4	57.5	(32)
(33)		10%	WB	52.6	53.7	55.4	57.2	60.1	63.8	65.0	64.7	63.5	59.7	56.3	53.5	(33)
(34)			MADB	55.6	58.2	62.1	67.3	73.5	80.6	81.9	81.6	80.8	71.7	61.7	55.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.7	20.6	22.4	24.1	26.7	29.7	29.9	30.0	31.6	29.2	23.3	19.3	(35)
(36)			MCDBR	27.2	29.9	32.3	34.2	36.9	39.5	38.9	40.0	42.9	40.2	31.6	24.7	(36)
(37)			MCWBR	16.8	17.3	17.6	17.6	17.5	17.0	17.0	16.8	17.5	17.2	17.0	15.6	(37)
(38)		5% WB	MCDBR	18.4	23.8	28.4	30.8	33.2	36.0	35.4	35.3	36.5	32.5	24.0	14.3	(38)
(39)			MCWBR	12.9	14.9	16.5	16.8	17.0	17.0	16.2	15.6	16.1	15.4	14.4	10.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.323	0.330	0.349	0.350	0.349	0.352	0.335	0.334	0.320	0.315	0.320	(40)	
(41)		taud	2.556	2.545	2.528	2.437	2.468	2.477	2.468	2.515	2.509	2.591	2.578	2.553	(41)	
(42)		Ebn at Noon	274	288	296	295	295	294	292	295	291	285	274	265	(42)	
(43)		Edh at Noon	24	27	30	35	35	35	35	32	30	25	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	717	1014	1438	1908	2268	2505	2405	2123	1777	1273	854	619	(44)	
(45)		RadStd	116	139	145	155	142	115	91	80	74	93	65	103	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +212	(m) N/A	(46)
(47)	Regional (39 neighbors)	N/A	N/A	-2.14	N/A	N/A	N/A	N/A	-187	+248	N/A	(47)

Nomenclature: See separate page